

5 Martingale Concentration Inequalities

5.1 Motivation and overview

Our goal is to get a tail bound for $X_1 + \dots + X_n$, where the X_i are independent. Here is our solution so far:

- (a) Chernoff inequality bounded by MGF.
- (b) Bound MGF using sub-Gaussian and sub-exponential properties.
- (c) Many commonly used random variables are sub-Gaussian or sub-exponential.

What about more complicated structure?

1. Sometimes, we want to show concentration of $S_n = f(X_1, \dots, X_n) =: f(X_{1:n})$.
2. Sometimes, we want to show concentration of $S_n = \sum_{t=1}^T X_t$, where $\{X_t\}_{t \geq 1}$ is correlated. We can deal with this if it is a Martingale difference sequence.

This lecture, we will take the approach of a Martingale concentration inequality. We will use Markov's inequality on $e^{\lambda S_n}$ along with a conditional MGF bound and optimizing over λ . We will see

- (a) Doob's Martingale representation
- (b) Azuma-Hoeffding, Azuma- Bernstein, and bounded difference inequalities
- (c) Applications
- (d) Variants: Freedman's inequality and Doob's maximal inequality

Example 5.1. Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_X \in \mathcal{P}([a, b])$. We want to estimate $\theta = \mathbb{E}_{X, X' \stackrel{\text{iid}}{\sim} P_X} [g(X, X')]$, where we assume that $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is symmetric (such as $g(x, x') = |x - x'|$ or $g(x, x') = \frac{1}{2}(x - x')^2$). In the latter case, $\theta = \text{Var}(X)$.

Hoeffding introduced ***U-statistics*** for estimating these parameters θ :

$$U(X_{1:n}) = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} g(X_i, X_j).$$

If we let

$$\hat{\mathbb{P}}_{X, X'} = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} \delta_{(X_i, X_j)}$$

be the empirical distribution, then $U(X_{1:n}) = \hat{E}_{(X, X')} [g(X, X')]$. The U statistic is an unbiased estimator of θ because

$$\mathbb{E}[U(X_{1:n})] = \mathbb{E}[g(X_i, X_j)] = \theta.$$

This has the smallest variance among all unbiased estimators.

Today, we will show the concentration bound

$$\mathbb{P}(|U - \theta| \geq t) \leq 2 \exp \left(-\frac{nt^2}{2\|g\|_\infty} \right).$$

This is significant because U is not a sum of independent random variables, so our previous technology does not work here.

5.2 Doob's martingale representation of $f(X_1, \dots, X_n)$

Now return to the setting where we are dealing with $f(X_1, \dots, X_n)$, where the X_i are independent. Define

$$Y_k = \mathbb{E}[f(X_{1:n}) \mid X_{1:k}] \quad k \geq 0.$$

We can think of conditioning on $X_{1:k}$ as conditioning on the σ -algebra $\mathcal{F}_k = \sigma(X_{1:k})$

Example 5.2. Here is the example to keep in mind: Let $f(X_{1:n}) = X_1 + \dots + X_n$ with independent X_i . Then

$$Y_k = X_1 + \dots + X_k + \mathbb{E}[X_{k+1}] + \dots + \mathbb{E}[X_n].$$

Further define the difference

$$D_k = Y_k - Y_{k-1}.$$

In the previous example, $D_k = X_k - \mathbb{E}[X_k]$. We can in general write

$$f(X) - \mathbb{E}[f(X)] = Y_n - Y_0 = \sum_{k=1}^n (Y_k - Y_{k-1}) = \sum_{k=1}^n D_k.$$

We call $\{Y_k\}$ a **martingale sequence** and $\{D_k\}$ a **martingale difference sequence**.

Let us recall what a martingale is.

Definition 5.1. A **filtration** is an increasing nested sequence of σ -algebras

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_n \subseteq \dots.$$

Often, we take $\mathcal{F}_k = \sigma(X_{1:k})$. If the filtration is not defined properly, the result you get may not be true.

Definition 5.2. If we have $\{Y_k\}_{k=1}^\infty$, where Y_k is $\mathcal{F}_k$ -measurable, then we say that $\{Y_k\}$ is **$\{\mathcal{F}_k\}$ -adapted**.

Definition 5.3. $\{(Y_k, \mathcal{F}_k)\}_{k \geq 1}$ is a **martingale sequence** if

1. $\{Y_k\}$ is adapted to $\{\mathcal{F}_k\}$.

2. $\mathbb{E}[|Y_k|] < \infty$,
3. $\mathbb{E}[Y_k \mid \mathcal{F}_{k-1}] = Y_{k-1}$.

Martingales are often used to model gambling problems where your strategy can depend on the outcomes of the past. If you don't have a martingale, you can sometimes subtract the mean to get one.

Definition 5.4. $\{D_k\}_{k \geq 1}$ is a **martingale difference sequence** if $\{\sum_{k=1}^n D_k\}_{n \geq 1}$ is a martingale with respect to $\{\mathcal{F}_k\}_{k \geq 1}$.

Example 5.3. Let $\{X_i\}_{i \geq 1} \stackrel{\text{iid}}{\sim} P_X$, where $\mathbb{E}[|X|] < \infty$. Denote $\mu = \mathbb{E}_X[X]$ and $S_k = \sum_{s=1}^k X_s$. Then $\{(X_k - k\mu, \sigma(X_{1:k}))\}_{k \geq 1}$ is a martingale.

Proof. We only need to check the third property:

$$\begin{aligned} \mathbb{E}[S_k - k\mu \mid X_{1:k-1}] &= S_{k-1} - (k-1)\mu \\ &= Y_{k-1}. \end{aligned} \quad \square$$

Example 5.4 (Doob's martingale). Let $\{X_i\}_{i \geq 1}$ be independent⁷ and $\mathbb{E}[|f(X_1, \dots, X_n)|] < \infty$. Then $\{(Y_k = \mathbb{E}[f(X_{1:n}) \mid X_{1:k}], \sigma(X_{1:k}))\}_{k \geq 1}$ is a martingale sequence.

Proof. Again, we only check the third property:

$$\begin{aligned} \mathbb{E}[Y_{k+1} \mid \sigma(X_{1:k})] &= \mathbb{E}[\mathbb{E}[f(X_{1:n}) \mid X_{1:n+1}] \mid X_{1:k}] \\ &= \mathbb{E}[f(X_{1:n}) \mid X_{1:k}] \\ &= Y_k \end{aligned}$$

The second equality is by the tower property of conditional expectation. $\square$

5.3 Martingale concentration

Most inequalities for an iid sum have a martingale version. Here is a martingale version of Bernstein's inequality.⁸

Theorem 5.1. Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence. If

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \leq e^{\lambda^2 \nu_k^2 / 2} \quad a.s. \quad \forall \lambda \leq \frac{1}{\alpha_k},$$

then

1. $\sum_{k=1}^n D_k$ is $\text{sE}(\sqrt{\sum_{k=1}^n \nu_k^2}, \max_{k \leq n} \alpha_k)$.

⁷In class, we had this assumption, but I don't think it is actually needed.

⁸This inequality does not have a formal name, but you may call it an Azuma-Bernstein inequality.

2.

$$\mathbb{P} \left(\left| \sum_{k=1}^n D_k \right| \geq t \right) \leq 2 \exp \left(- \min \left\{ \frac{t^2}{2 \sum_{k=1}^n \nu_k^2}, \frac{t}{2\alpha_*} \right\} \right)$$

This condition is that a random variable given by the MGF is bounded. We will see later how to check this condition.

Proof. We can start with the Chernoff bound

$$\mathbb{P} \left(\sum_{k=1}^n D_k \geq t \right) \leq \inf_{\lambda} \frac{\mathbb{E}[e^{\lambda \sum_{k=1}^n D_k}]}{e^{\lambda t}}.$$

Then we can bound the moment generating function by using the tower property of conditional expectation

$$\mathbb{E}[e^{\lambda \sum_{k=1}^n D_k}] = \mathbb{E}[e^{\lambda \sum_{k=1}^{n-1} D_k} \mathbb{E}[e^{\lambda D_n} \mid \mathcal{F}_{n-1}]]$$

Using $\lambda \leq \frac{1}{\alpha_n}$,

$$\begin{aligned} &\leq \mathbb{E}[e^{\lambda \sum_{k=1}^{n-1} D_k} e^{\lambda^2 \nu_k^2 / 2}] \\ &= \mathbb{E}[e^{\lambda \sum_{k=1}^{n-1} D_k}] e^{\lambda^2 \nu_k^2 / 2} \end{aligned}$$

Iterating this argument, we get

$$\leq e^{\lambda^2 (\sum_{k=1}^n \nu_k^2) / 2}$$

for all $\lambda \leq \frac{1}{\max_{k \leq n} \alpha_k}$. □

Remark 5.1. In this theorem, the ν_k are deterministic. In the case where the ν_k are $\mathcal{F}_{k-1}$ -measurable, we will get a related but different bound.

Here is a corollary which is sometimes easier to use than the previous theorem.

Corollary 5.1 (Azuma-Hoeffding inequality). *Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence. Suppose there exists $\{(a_k, b_k)\}_{k=1}^n$ such that $D_k \in (a_k, b_k)$ a.s., where b_k, a_k are $\mathcal{F}_{k-1}$ -measurable and $|b_k - a_k| \leq L_k$. Then*

$$1. \sum_{k=1}^n D_k \text{ is sG}(\sqrt{\sum_{k=1}^n L_k^2 / 2}).$$

2.

$$\mathbb{P} \left(\left| \sum_{k=1}^n D_k \right| \geq t \right) \leq 2 \exp \left(- \frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2} \right).$$

Proof. We have $\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \leq e^{\lambda^2 (b_k - a_k)^2 / 8}$. Use the same proof as before. □

Now specialize to Doob's martingale

$$D_k = \mathbb{E}[f(X_{1:n}) \mid X_{1:k}] - \mathbb{E}[f(X_{1:n}) \mid X_{1:k-1}].$$

Definition 5.5. $f(x_1, \dots, x_n)$ is a **bounded difference function** if for all $k \in [n]$, $x_{1:n}, x'_k$,

$$|f(x_{1:k-1}, x_k, x_{k+1:n}) - f(x_{1:k-1}, x'_k, x_{k+1:n})| \leq L_k.$$

This is a condition on how much the function changes if we change 1 coordinate. Here is a corollary of the Azuma-Hoeffding inequality

Corollary 5.2. Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is $L_{1:n}$ bounded and $X_{1:n}$ has independent components. Then for all $t \geq 0$,

$$\mathbb{P}(|f(X_{1:n}) - \mathbb{E}[f(X_{1:n})]| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n L_k^2}\right).$$

Proof. This is Azuma-Hoeffding with $\sum_{k=1}^n D_k = f(X_{1:n}) - \mathbb{E}[f(X_{1:n})]$. Here, there exist $A_k \leq D_k \leq B_k$, where $|B_k - A_k| \leq L_k$ because we can let

$$B_k = \sup_x \mathbb{E}[f(X_{1:n}) \mid X_{1:k-1}, X_k = x] - \mathbb{E}[f(X_{1:n}) \mid X_{1:k-1}],$$

$$A_k = \inf_x \mathbb{E}[f(X_{1:n}) \mid X_{1:k-1}, X_k = x] - \mathbb{E}[f(X_{1:n}) \mid X_{1:k-1}]. \quad \square$$

5.4 Applications

Example 5.5 (U -statistics). Here is how we can get a concentration inequality for U -statistics: Recall that

$$U(X_{1:n}) = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} |X_i - X_j|, \quad X_i \sim P_X \in \mathcal{P}([-b, b]).$$

Then

$$\begin{aligned} |U(X_{1:k-1}, X_k, X_{k+1:n}) - U(X_{1:k-1}, X'_k, X_{k+1:n})| &= \frac{1}{\binom{n}{2}} \left| \sum_{s \neq k} |X_s - X_k| - |X_s - X'_k| \right| \\ &\leq \frac{1}{\binom{n}{2}} \sum_{s \neq k} |X_k - X'_k| \\ &\leq \frac{2}{n(n-1)} \cdot (n-1) \cdot 2b \\ &\leq \frac{4b}{n}. \end{aligned}$$

So U is $(\frac{4b}{n}, \frac{4b}{n}, \dots, \frac{4b}{n})$ -bounded difference. This gives the tail bound

$$\mathbb{P}(|U(X_{1:n}) - \theta| \geq t) \leq 2 \exp\left(-\frac{2t^2}{n \frac{16}{n^2}}\right) = 2 \exp\left(-\frac{nt^2}{16}\right).$$

That is,

$$|U(X_{1:n}) - \theta| \lesssim b \sqrt{\frac{\log(2/\delta)}{n}} \quad \text{with probability } 1 - \delta.$$

Example 5.6 (Supremum of empirical process). Suppose we have samples $(Z_i)_{i \in [n]} \stackrel{\text{iid}}{\sim} P_Z$, where $Z_i = (X_i, Y_i)$. We can define the **loss function** $\ell : Z \times \Theta \rightarrow [0, 1]$ and the **empirical risk**

$$\hat{R}_n(\theta) = \frac{1}{n} \sum_{k=1}^n \ell(Z_k; \theta).$$

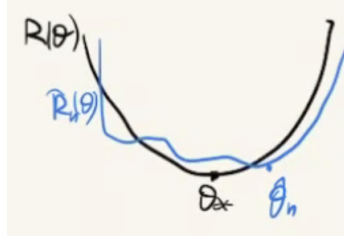
Correspondingly, we have the **population risk**

$$R(\theta) = \mathbb{E}[\hat{R}_n \mid \theta] = \mathbb{E}[\ell(Z; \theta)]$$

In statistical learning theory, we are often concerned with the **excess risk**

$$\mathcal{E}[Z_{1:n}] := \sup_{\theta \in \Theta} R(\theta) - \hat{R}_n(\theta).$$

We can use an **empirical risk minimizer** $\hat{\theta}_n$, and we want to upper bound $R(\hat{\theta}_n) \leq \hat{R}_n(\hat{\theta}_n) + \mathcal{E}(Z_{1:n})$.



We claim that $\mathcal{E}(Z_{1:n})$ is $(1/n, \dots, 1/n)$ -bounded difference. Then

$$|\mathcal{E}(Z_{1:n}) - \mathbb{E}[\mathcal{E}(Z_{1:n})]| \leq \sqrt{\frac{\log(2/\delta)}{2n}} \quad \text{with probability } 1 - \delta.$$

Proof. Fix $Z_{1:n}$, and let $\theta_* = \arg \max_{\theta \in \Theta} (R(\theta) - \hat{R}_n(\theta))$. Then $\mathcal{E}(Z_{1:n}) = R(\theta_*) - \hat{R}_n(\theta_*)$. We want to look at

$$|\mathcal{E}(Z_{1:n}) - \mathcal{E}(Z_{1:k-1}, Z'_k, Z_{k+1:n})| = \frac{1}{n} \sum_{i=1}^n (\ell(Z_i; \theta_*) - \mathbb{E}[\ell(Z_i; \theta_*)])$$

$$\begin{aligned}
& - \sup_{\theta \in \Theta} \frac{1}{n} \sum_{i \neq k} (\ell(Z_i; \theta) - \mathbb{E}[\ell(Z_i; \theta)]) \\
& - \frac{1}{n} (\ell(Z'_k; \theta) - \mathbb{E}[\ell(Z'_k; \theta)]) \\
& \leq \frac{1}{n} \sum_{i=1}^n (\ell(Z_i; \theta_*) - \mathbb{E}[\ell(Z_i; \theta_*)]) \\
& \quad - \frac{1}{n} \sum_{i \neq k} (\ell(Z_i; \theta_*) - \mathbb{E}[\ell(Z_i; \theta_*)]) \\
& \quad - \frac{1}{n} (\ell(Z'_k; \theta_*) - \mathbb{E}[\ell(Z'_k; \theta_*)]) \\
& = \frac{1}{n} (\ell(Z_k; \theta_*) - \ell(Z'_k; \theta_*)) \\
& \leq \frac{1}{n}.
\end{aligned}$$

□

Remark 5.2. This doesn't say anything about

$$\mathbb{E} \left[\sup_{\theta} \hat{R}_n(\theta) - R(\theta) \right].$$

5.5 Freedman's inequality

Our “Azuma-Bernstein” inequality says that if $\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \leq e^{\lambda^2 \nu_k^2 / 2}$, then

$$\left| \frac{1}{n} \sum_{k=1}^n D_k \right| \leq \max \left\{ \sqrt{\frac{\frac{2}{n} \sum_{k=1}^n \nu_k^2}{n} \log \left(\frac{2}{\delta} \right)}, \frac{2\alpha_* \log \left(\frac{2}{\delta} \right)}{n} \right\} \quad \text{with probability } 1 - \delta.$$

However, sometimes ν_k^2 is not deterministic and instead is $\mathcal{F}_{k-1}$ measurable.

Theorem 5.2 (Freedman's inequality). *Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence such that*

1. $\mathbb{E}[D_k \mid \mathcal{F}_{k-1}] = 0$.
2. $D_k \leq b$ a.s.

Then for all $\lambda \in (0, 1/b)$ and $\delta \in (0, 1)$,

$$\mathbb{P} \left(\sum_{t=1}^T X_t \leq \lambda \sum_{t=1}^T \mathbb{E}[D_k^2 \mid \mathcal{F}_{k-1}] + \frac{\log(1/\delta)}{\lambda} \right) \geq 1 - \delta.$$

This is useful in bandit and reinforcement learning research.⁹

⁹For example, see Theorem 1 in Beygelzimer, Langford, et. al. 2010.

5.6 Maximal Azuma-Hoeffding inequality

Recall Doob's maximal inequality for sub-martingales.

Lemma 5.1 (Doob's maximal inequality). *If $\{X_s\}_{s \geq 0}$ is a sub-martingale, i.e.*

$$X_s \leq \mathbb{E}[X_t \mid \mathcal{F}_s] \quad \forall s < t,$$

then for all $u > 0$,

$$\mathbb{P} \left(\sup_{0 \leq t \leq T} X_t \geq u \right) \leq \frac{\mathbb{E}[\max\{X_T, 0\}]}{u}.$$

This gives rise to a maximal version of the Azuma-Hoeffding inequality:

Theorem 5.3 (Maximal Azuma-Hoeffding inequality). *Let $\{(D_k, \mathcal{F}_k)\}$ be a martingale difference sequence, and suppose there exists $\{(a_k, b_k)\}_{k=1}^n$ such that $D_k \in (a_k, b_k)$ a.s. Then*

$$\mathbb{P} \left(\sup_{0 \leq k \leq n} \sum_{s=1}^k D_s \geq t \right) \leq \exp \left(- \frac{2t^2}{\sum_{k=1}^n (b_k - a_k)^2} \right).$$